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Q.1 ABC is a network of Hospitals. Its having Hospitals across the World. Its possible to take appointment for any Doctor for an available slot on an available date through its website. Its having a Chatbot also. There are many specialities in each of its Hospitals. However, not all specialities are available across all its Hospitals. Doctors may provide consultancy services in more than one hospital of its network. Its possible to make payment of consultation fee online while taking appointment or directly at hospital prior to consultation.

Make assumptions, wherever necessary. Also, list them very clearly.

Now, with reference to the above, answer the following:

- (1) List the Entities**
- (2) For each Entity, list Attributes**
- (3) Define relationships between the Entities**
- (4) Finally, draw the Entity Relationship Diagram**

Ans. (1) List the Entities

ABC Hospital ER diagram illustrates key information about Hospital, including entities such as doctor, patient, specialization and chatbot etc. It allows to understand the relationships between entities.

- Hospital
- Doctor
- Patient
- Specialization
- Appointment
- Payment
- Chatbot

Table: Hospital

Field Name	Data Type	Constraint	Description
Hospital_id	int	Primary Key	Uniquely identify each hospital in this table
Hospital_name	varchar(100)		Name of the hospital
Location	varchar(200)		Location of the hospital
Contact no.	int		Doctor's contact details

Table: Doctor

Field Name	Data Type	Constraint	Description
Doctor_id	int	Primary Key	Uniquely identify each Doctor in this table
Doctor_name	varchar(50)		Name of the doctor

Gender	varchar(100)		Gender of the doctor
Contact no.	varchar(50)		Contact no. of the doctor

Table: Patient

Field Name	Data Type	Constraint	Description
Patient_id	int	Primary Key	Uniquely identify each Patient in this table
Patient_name	varchar(50)		Name of the Patient
Contact_no.	varchar(100)		Contact no. of the patient
Mail_id	varchar(50)		Mail id of the patient

Table: Specialization

Field Name	Data Type	Constraint	Description
Specialization_id	int	Primary Key	Uniquely identify each Specialization in this table
Specialization_name	varchar(50)		Name of the Specialization
Description	varchar(100)		Description of the Specialization

Table: Appointment

Field Name	Data Type	Constraint	Description
Appointment_id	int	Primary Key	Uniquely identify each Appointment in this table
Patient_id	varchar(50)	Foreign Key	identify each Patient in this table
Doctor_id	varchar(100)		identify each Doctor in this table
Hospital_id	varchar(50)		identify each hospital in this table
Date	Date and Time		Date and time of Appointment

Table: Payment

Field Name	Data Type	Constraint	Description
Payment_id	int	Primary Key	Uniquely identify each Payment in this table
Appointment_id	varchar(50)	Foreign Key	identify each Appointment in this table
Payment_Mode	varchar(100)		Mode of Payment
Payment_Status	varchar(50)		Status of Payment
Amount	varchar(50)		Amount to pay

Table: Chatbot

Field Name	Data Type	Constraint	Description
Chatbot_id	int	Primary Key	Uniquely identify each Chatbot in this table
Functionality	varchar(50)		Functionality of chat patient
Availability	varchar(100)		Available for chat

(2) For each Entity, list Attributes

- **Hospital Entity** : Attributes of hospital Entity are Hospital_id, Hospital_name, Location, Contact no., id is Primary Key for hospital Entity.
- **Doctor Entity** : Attributes of hospital Entity are Doctor_id, Doctor_name, Gender, Contact no. Doctor_id is Primary Key for doctor Entity.

- **Patient Entity** : Attributes of patient Entity are Patient_id, Patient_name, Contact_no., Mail_id. Patient id is Primary Key for Patient Entity.
- **Specialization Entity** : Attributes of Specialization Entity are Specialization_id, Specialization_name, Description. Specialization id is Primary Key for Specialization Entity.
- **Appointment Entity**: Attributes of appointment Entity are Appointment_id, Patient_id, Doctor_id, Hospital_id, Date. Appointment id is Primary Key for Appointment Entity.
- **Payment Entity**: Attributes of Payment Entity are Payment_id, Appointment_id, Payment_Mode, Payment_Status, Amount. Payment id is Primary Key for Payment Entity.
- **Chatbot Entity**: Attributes of Chatbot Entity are Chatbot_id, Functionality, Availability. Chatbot id is Primary Key for Chatbot Entity.

Attributes: Hospital

Hospital ID	Hospital Name	Location	Contact No.
101	Mass	Delhi	5471425417
102	Manipal	Jaipur	3521475474
103	Metro	Jaipur	6521452111
104	Saket	Delhi	6542154747
105	Shine	Mumbai	5654474521

Attributes: Doctor

Doctor id	Doctor Name	Gender	Contact No.
201	Sunil Poonia	Male	4785478741
202	Sonam	Female	6584214541
203	Sandeep	Male	3254124512
204	Priyanka Payal	Female	5632145214
205	Pradeep	Male	4521452141

Attributes: Patient

Patient id	Patient Name	Contact No.	Mail Id
301	Rakesh	5475214744	Rakesh66@gmail.com
302	Pawan	5414784521	Pawan99@gmail.com
303	Mohan	2321474541	Mohan89@gmail.com
305	Sandeep	8745471474	Sandeep987@gmail.com
306	Sonia	5654742147	Sonia234@gmail.com

Attributes: Specialization

Specialization id	Specialization Name	Description
401	Heart	Doctor of heart
402	Leaver	Doctor of leaver
403	Ear	Doctor of ear
404	Eye	Doctor of eye
405	Heart	Doctor of heart

Appointment id	Patient id	Doctor id	Hospital id	Date
501	301	201	101	02/03/2025
502	302	202	102	05/04/2025
503	303	203	103	10/03/2025
504	305	204	104	09/02/2025
505	306	205	105	25/04/2025

Attributes: Payment

Payment id	Appointment id	Payment Mode	Payment Status	Payment
601	501	Online	Successful	201000
602	502	Online	Successful	50000
603	503	Offline	Successful	75000
604	504	Online	Successful	25000
605	505	Offline	Successful	10000

Attributes: Chatbot

Chatbot id	Functionality	Availability
701	Yes	Available
702	Yes	Available
703	No	Not Available
704	Yes	Available
705	Yes	Available

(3) Define relationships between the Entities

Relationships are:

- Hospital - Doctor:** A hospital can have many doctors, and a doctor can be associated with multiple hospitals.
Relationship Type: Many-to-Many.
- Doctor - Specialization:** A doctor belongs to one specialization, but a specialization can have many doctors.
Relationship Type: One-to-Many.
- Patient - Appointment:** A patient can book multiple appointments, but each appointment is for one patient.
Relationship Type: One-to-Many.
- Appointment - Doctor:** An appointment is for one doctor, but a doctor can have multiple appointments.
Relationship Type: One-to-Many.
- Appointment - Hospital:** An appointment is made for a doctor at a specific hospital.
Relationship Type: Many-to-One.
- Appointment - Payment:** Each appointment can have one payment, but each payment is linked to one appointment.
Relationship Type: One-to-One.
- Chatbot - Patient:** The chatbot can assist many patients, but it is a general-purpose tool.
Relationship Type: One-to-Many.

(4) Finally, draw the Entity Relationship Diagram

ER Diagram is known as Entity-Relationship Diagram, it is used to analyze the structure of the Database. It shows relationships between entities and their attributes. An ER Model provides a means of communication.

The ER diagram for the hospital network has been created:

